

Metric-Semantic Primitive Matching for Cross-View Robot Localization Beyond Urban Environments

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Abstract—Successful robot automation requires accurate localization in a global coordinate frame for repeatability, task planning and goal specification, and safe operation. Still, global localization in GNSS-denied settings is an open problem in robotics. Overhead aerial imagery offers an excellent candidate for accomplishing this objective. However, current methods focus on localizing in structured urban areas and have shown limited application in unstructured, natural environments. Limitations to the state-of-the-art include a reliance on pre-training learned models for operation in specific areas and difficulty dealing with repetitive geometries and featureless landscapes found in natural outdoor areas. To this end, we present a method for extracting and matching high-level metric-semantic primitives across aerial images and ground robot RGB-D camera data that we show generalizes well across many environments. Our method operates in a zero-shot manner, requiring no training on area-specific data and only an aerial orthophoto at runtime. We demonstrate our algorithm can localize a ground robot across a wide variety of environments, including an autonomous driving dataset, a park area, and a wilderness camp, at an average error of 2.7 m when operating on an out-of-date aerial overhead image, and 2.0 m when operating on recent aerial imagery.

I. INTRODUCTION

Consistent robot localization in a global coordinate frame is a foundational prerequisite to downstream autonomy tasks [1]. Aerial imagery acquired from satellites [2, 3] or drones [4] has shown significant utility for consistent pose estimation. Large-scale aerial images can serve as a globally consistent reference for a ground robot, enabling it to match features in its local sensor frame with features in the aerial image and estimate its absolute pose in a meaningful global coordinate frame. Furthermore, *cross-view localization*—the task of estimating a ground pose from an overhead image—enables global pose estimation without requiring a robot to revisit a location (e.g. loop closure in a SLAM system) or use a prior map, which can be costly to acquire and maintain [5].

Localizing in an aerial image offers many benefits in downstream autonomy as well. Knowing a robot pose in an aerial image gives a human operator easy situational awareness and can serve as a means for the operator to specify mission plans and for the robot to communicate results of performing tasks [6]. Cross-view localization can be useful for multi-robot

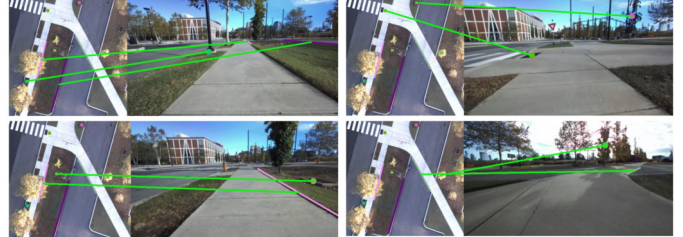


Fig. 1: Our algorithm maps and matches object points and line edges across aerial and ground views, intuitively representing abstract elements of an environment that are key indicators for localization. We propose an efficient and robust data association method that leverages semantic and geometric cues to find correct associations, enabling cross-view localization without any environment-specific training.

collaboration as well, enabling robots to share maps and plans without requiring robots to ever be near each other.

The utility offered by cross-view localization has made it a well-studied problem in urban domains; however, cross-view localization in unstructured natural environments has been under-explored. Instead, most state-of-the-art cross-view localization methods focus on self-driving applications, primarily in urban environments where dense buildings and road structures provide strong geometric guidance [3, 7–9]. Localizing from aerial imagery in forested or mountainous environments brings significant challenges, including repetitive geometries or featureless landscapes, which can result in localization difficulty. Furthermore, most current cross-view localization work relies on a network that has been trained in similar areas to the region of operation [7, 10], and suffers from an inability to bridge the domain gap when operating in novel regions.

In contrast, our work presents a method for performing cross-view localization in general environments and does not rely on any prior training, enabling easy generalization. We leverage foundational segmentation models [11, 12] and semantic models [13] to extract high-level metric-semantic primitives from aerial imagery and ground trajectories. Using semantic similarity and geometric consistency, we solve an outlier-robust data association problem to match primitives across views (shown in Fig. 1) and solve for instantaneous ground robot pose. We combine periodic registration measurements with an outlier rejection mechanism to perform robust pose graph optimization over the course of a robot trajectory and demonstrate highly accurate zero-shot pose estimation without any environment-specific training. Furthermore, we

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demonstrate our method’s ability to perform cross-view localization in a variety of environments, including a self-collected, semi-unstructured dataset.

II. RELATED WORK

Research in cross-view localization spans both coarse, region-level position estimation (on the order of tens or hundreds of meters) and meter or sub-meter-level fine-grained pose estimation.

Coarse Cross-View Localization. Early cross-view localization work formulated air-ground localization as a database query problem, where the goal was to retrieve the best aerial image from a database given a query ground image [14]. While early cross-view works focused on computing hand-crafted descriptors for retrieving corresponding aerial images, recently, using Siamese networks to learn global descriptors for ground images and aerial images in a joint embedding space has shown impressive aerial position retrieval capabilities [15, 16].

Fine-Grained Cross-View Pose Estimation. Recent advances in learned architectures and training schemes have enabled neural networks to perform fine-grained, meter-level pose estimation from ground images [10]. Follow-up works investigate hybrid learning and matching approaches [7] and leveraging domain priors, such as the understanding that higher elevation features tend to be more visible from overhead imagery [9].

Cross-View Robot Trajectory Estimation. Estimating a robot’s full trajectory in an aerial image extends fine-grained cross-view pose estimation to trajectory estimation over time. Approaches to this problem include both incorporating likelihood measurements of potential poses into a particle filter [2, 4] and incorporating fine-grained pose measurements into a factor graph [8]. Miller et al. [4] propose an algorithm for extracting coarse semantics from a LiDAR point cloud and efficiently score potential poses based on polar class-wise images. Most similar to our work, Ankenbauer et al. [17] detect and match predefined objects (rocks and cars) across views. Adjacent to cross-view localization, [18–20] use OpenStreetMaps (OSM) for building and road constraints for accurate localization. In comparison, we use only a single limited-field-of-view camera with depth values (which can also be obtained from LiDAR-RGB fusion) and an aerial orthophoto, requiring no scene-specific training or prior map.

III. PROPOSED APPROACH

We propose a method for estimating the trajectory of a robot in an aerial image using a single limited-field-of-view camera with depth measurements. The fundamental idea behind this method is that abstract, high-level point and line features can be used as a sparse environment representation shared between both aerial and ground views. By matching these sparse primitives, robot poses can be tracked across a trajectory of camera inputs. A diagram of our cross-view mapping, matching, and trajectory estimation pipeline is shown in Fig. 2, broken up into three submodules: Mapping (Section III-A),

Matching and Registration (Section III-B), and Robust Pose Graph Optimization (Section III-C).

During mapping, sparse points and lines with semantic descriptors are extracted from aerial image patches and ground RGB-D data. At regular distances, the ground robot creates submaps of nearby points and lines. A place recognition module is used to recommend aerial patches for registration, and then our graph-theoretic data association algorithm is used to find points and lines that are common between ground and aerial views. Matches are used to extract a transformation between the center of a submap and the associated aerial patch. These transformations are supplied to the robust pose graph optimization, which rejects outlier transformations and estimates the robot trajectory. Once the robot’s rough global pose has been determined, a second, minimal round of matching, registration, and pose graph optimization is performed using the pose initialization to refine the matching solution.

A. Ground Primitive Mapping and Aerial Primitive Extraction

The central goal of our sparse primitive extraction approach is to find a set of sparse, high-level features represented commonly in the aerial and ground views. To do this, we build off our prior work, ROMAN [21], to extract centroid-based objects from aerial and ground views (*i.e.* point features). We extend this approach by additionally extracting 2D lines to assist with matching in areas where there are insufficient point features seen in both views.

To extract aerial points and lines, we first split an aerial reference image into overlapping aerial patches and run Segment Anything [12] on each aerial patch. Each small segment whose area is below a threshold is represented as a single point feature defined by the segment centroid. Alpha shapes are then used to extract the border of each large segment, and fine-grained borders are fused together into coarse, straight lines when suitable. Additionally, we run DINOv2 [13] on images, interpolate the output token embeddings, and assign each primitive a semantic embedding based on the DINO semantics. Since the aerial imagery is typically obtained before a ground robot experiment, this computation can be done offline beforehand.

Ground metric-semantic primitives are created by first mapping segments in 3D with ROMAN [21]. Local submaps are created at distance-defined intervals, with 3D segments projected to 2D and converted to sparse points and lines following the same procedure as aerial primitive extraction. The most significant challenge to creating clean point/line maps is to reject lines that are a result of a limited field-of-view (FOV). For example, the edge of the camera frustum could easily be converted to a line, over-cluttering the 2D segment map. To overcome this challenge, while mapping we keep track of *FOV borders*. Given an initial set of input segments, points along the edge of the camera and points at the maximum depth range are marked as FOV border points. When converting to 2D sparse primitives, lines are only created when a significant portion of that line is sufficiently

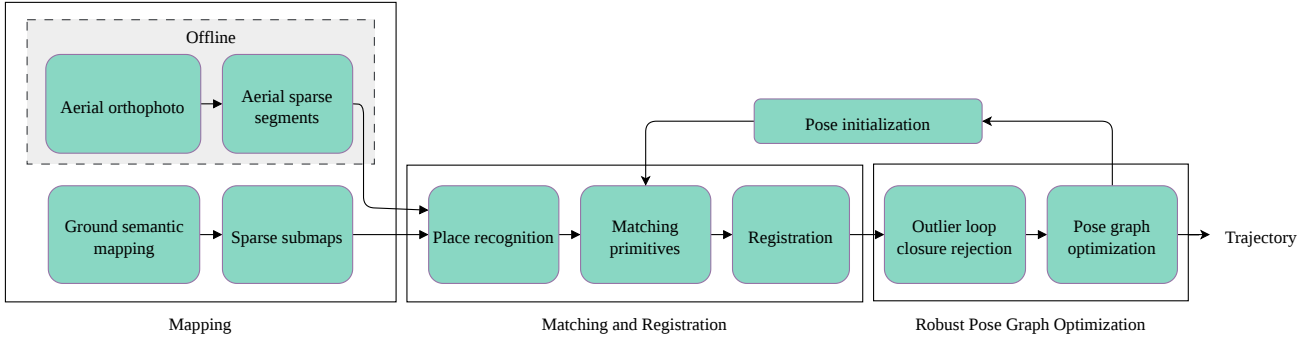


Fig. 2: Our cross-view localization pipeline includes three modules. First, sparse primitives are extracted and grouped into aerial patches and ground submaps. Submaps and aerial patches are fed to the matching and registration module for detecting cross-view loop closures. Then, pose graph optimization is performed to estimate the ground trajectory. Matching is re-run with the resulting pose constraining the rotation to refine the trajectory estimate. We run mapping at 6 Hz. Submap creation and matching are triggered every 20 m for ground robots and every 40 m for the faster-moving KITTI vehicle.

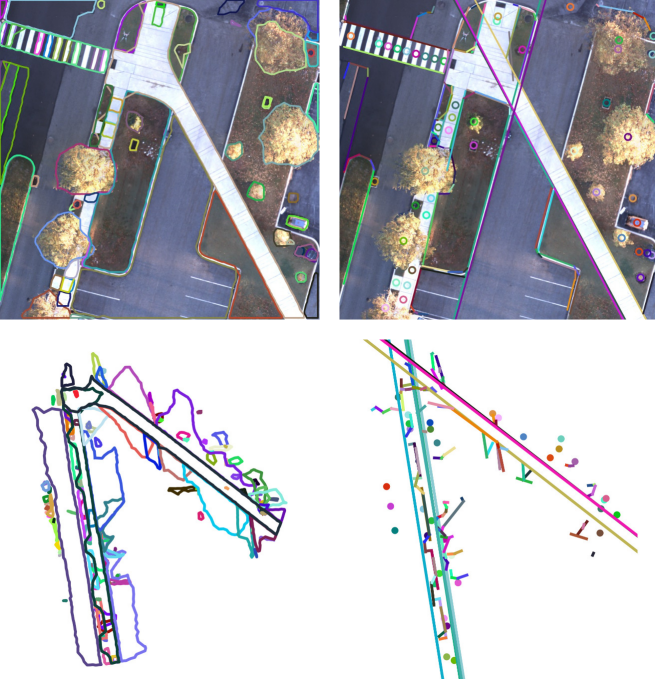


Fig. 3: Example segment maps and extracted primitives (points and lines). Clockwise from top left: (1) aerial segment map, (2) aerial primitives, (3) ground primitives, and (4) ground segment map. This figure illustrates one aerial patch and ground submap pair from our self-collected Park + Meadow dataset.

distant from FOV border points. Example aerial and ground segment maps and primitives are shown in Fig. 3.

B. Metric-Semantic Primitive Matching

Given a ground submap and set of aerial patches, we select a subset of aerial patches on which to attempt primitive matching and registration. We use AnyLoc [22] to create descriptors for each aerial patch and for regularly sampled poses along the ground submap trajectory. The subset of similar aerial patches is passed to the matching module to compute primitive correspondences.

Within the matching module, given a set of aerial points and

lines (p^a, ℓ^a) and ground points and lines (p^g, ℓ^g) , we construct a set of putative associations based off the k most semantically similar primitives. Each association $a_i = (o^a, o^g)$ is between two points or two lines, with superscripts a and g denoting primitives in the aerial and ground frames, respectively. Given these putative associations, we construct an affinity matrix, $\mathbf{M} \in \mathbb{R}^{n \times n}$ where $\mathbf{M}(i, j) = s(a_i, a_j) \in [0, 1]$ for all i, j , and where $s(a_i, a_j)$ is a weighted consistency scoring function between a pair of associations a_i and a_j . Here, a higher consistency score denotes a pair of associations that are relatively more consistent. We find a robust set of inlier correspondences by solving

$$\max_{\mathbf{u} \in \{0,1\}^n} \frac{\mathbf{u}^\top \mathbf{M} \mathbf{u}}{\mathbf{u}^\top \mathbf{u}}. \quad (1)$$

$$\text{subject to } \mathbf{u}_i \mathbf{u}_j = 0 \text{ if } \mathbf{M}_{i,j} = 0, \forall i, j.$$

Rather than estimating a single data association solution, we use [23] to solve for a distribution of data associations, which is modeled as a set of particles. We select the top n most frequent particles to be used as data association hypotheses.

Importantly, we propose a novel distance metric for points and lines that differentiates between infinite lines and bounded rays and line segments. Existing approaches for line associations use the affine Grassmanian metric [24], which treats lines as affine subspaces. This is limiting for two reasons. First, since all non-parallel infinite 2D lines intersect, the Grassmanian metric between lines reduces to only the angle between two lines, which does not discriminate enough between correct and spurious matches in 2D. Second, it is not clear how to incorporate known rotation or translation priors into the Grassmanian metric, which we show in Table III is helpful when a coarse rotation or translation estimate is available.

We instead propose a scoring metric that is general to both infinite and bounded lines and allows incorporation of a known rotation constraint. We define

$$s_{\text{pair}}(a_i, a_j) = \begin{cases} \text{GM}(s_d(a_i, a_j), s_p(a_i, a_j)), & \text{is_line}(a_i, a_j) \\ s_p(a_i, a_j), & \text{otherwise,} \end{cases} \quad (2)$$

where $\text{GM}(\cdot)$ denotes the geometric mean.

We define the directional scoring function to be

$$s_d(a_i, a_j) = \exp\left(\frac{-|\text{angle}(o_i^a, o_j^a) - \text{angle}(o_i^g, o_j^g)|}{2\sigma_d^2}\right), \quad (3)$$

where $\text{angle}(o_i, o_j) = \tan^{-1}(|\mathbf{d}(o_i)^\top \mathbf{d}(o_j)|)$ measures the angle between two line directions and $\mathbf{d}(o_i)$ is the direction vector of a line o_i . By taking the absolute value, we ensure that we take the smaller of the two angles formed by two lines. Then, $s_d(a_i, a_j)$ measures the difference between the two angles formed by the pair of lines with a weighting noise term σ_d . The two angles should be invariant between datasets if the two associations a_i and a_j are consistent.

The position-based invariant $s_p(a_i, a_j)$ scores the similarity between two point or line objects. Let

$$\mathbf{c}(a_i, a_j) = \tilde{\mathbf{p}}_i(o_i, o_j) - \tilde{\mathbf{p}}_j(o_i, o_j) \quad (4)$$

where $\tilde{\mathbf{p}}_i(o_i, o_j) \in \mathbb{R}^3$ returns the point on primitive o_i that is nearest to object o_j . So, $\mathbf{c}$ defines a vector between the two closest points on objects o_i and o_j . Then, we define the position-based invariant

$$s_p(a_i, a_j) = \exp\left(\frac{-\|\mathbf{c}^a\| - \|\mathbf{c}^g\|}{2\sigma_p^2}\right). \quad (5)$$

The key insight here is that the distance between the two closest points on two objects is invariant to rotation and translation, so we can use this distance as a consistency check on pairs of associations.

Furthermore, if a prior on rotation exists (*e.g.*, from a magnetometer), s_p can be split into x and y components, diminishing the effect of incorrect, high objective matches that result in an incorrect rotation.

We follow our findings in [21] and incorporate semantics into the overall score

$$s(a_i, a_j) = \text{GM}(s_{\text{pair}}(a_i, a_j), s_{\text{sem}}(a_i), s_{\text{sem}}(a_j)) \quad (6)$$

where s_{sem} is the scaled cosine similarity of the semantic descriptors of two associated primitives.

Once the affinity matrix $\mathbf{M}$ is constructed, the distribution over potential loop closures is estimated using [23], which is represented as a set of particles where each particle represents a set of putative associations. For each set of putative associations, a 2D transformation between the center of the aerial patch and the center of the ground submap is computed using an altered Arun's method [25]. Line directions are treated equivalently to centroid-adjusted points when constructing the cross-covariance matrix, ensuring that the rotation aligns both points and line directions. Translation is estimated by jointly minimizing point and line representation errors via weighted least squares. To resolve line direction ambiguities, a minimal non-degenerate subset of associated points and/or lines is selected, and all possible line direction combinations are enumerated, with a transform computed for each. The transformation yielding the smallest error across all primitives is used to fix line directions, after which a refined transform is

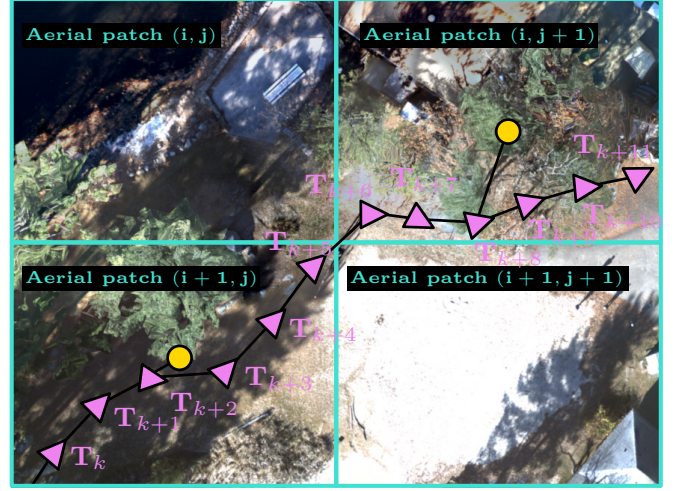


Fig. 4: Cross-view pose graph. Odometry edges encode the transformation between ground submap poses and cross-view registration measurements form edges between ground poses and aerial patch centers.

computed using all inlier associations. The particles are then clustered using the relative transform they yield and n loop closures are extracted using the top n clusters ($n = 25$ is used throughout the experiments). Outlier rejection is then performed with CLIPPER [26] to extract a single set of consistent loop closures (*i.e.* inlier loop closures).

C. Robust Pose Graph Optimization

We formulate cross-view trajectory estimation as a robust pose graph optimization problem where the graph consists of ground nodes $\mathbf{T}_k$, $k = 1, \dots, K$, representing the K ground poses, and aerial nodes $\hat{\mathbf{T}}_{i,j}^a$, $i = 1, \dots, I$, $j = 1, \dots, J$, representing an aerial image divided into $I \times J$ patches. Since aerial patches are already in the global frame, we only optimize the ground poses. We write a cross-view registration measurement as $\mathbf{T}_{\text{ground}_k}^{\text{aerial}_{i,j}}$ where (i, j) denotes the aerial patch and k denotes the ground pose index. Because place recognition in cross-view regimes is difficult, this graph has many outliers. Inspired by [27], we use a similar dense clique finding technique to the data association optimization in Equation (1), where each element of the affinity matrix $\mathbf{M}$ represents consistency between a pair of loop closures rather than a pair of associations. For example, in the case of Fig. 4, the consistency score between the loop closure $\hat{\mathbf{T}}_{\text{ground}_{k+2}}^{\text{aerial}_{i+1,j}}$ and $\hat{\mathbf{T}}_{\text{ground}_{k+8}}^{\text{aerial}_{i,j+1}}$ would be high if the difference between $\mathbf{T}_{\text{aerial}_{i,j+1}}^{\text{aerial}_{i+1,j}}$ and $\hat{\mathbf{T}}_{\text{ground}_{k+2}}^{\text{aerial}_{i+1,j}} \mathbf{T}_{\text{ground}_{k+8}}^{\text{ground}_{k+2}} \left(\hat{\mathbf{T}}_{\text{ground}_{k+8}}^{\text{aerial}_{i,j+1}} \right)^{-1}$ are similar. After constructing the affinity matrix, CLIPPER [26] is used to detect inlier loop closures. Outlier loop closures are then removed from the pose graph before trajectory optimization is performed to estimate the robot's trajectory in the global frame.

IV. EXPERIMENTS

To evaluate our full cross-view localization pipeline, we show the performance of our algorithm in several different

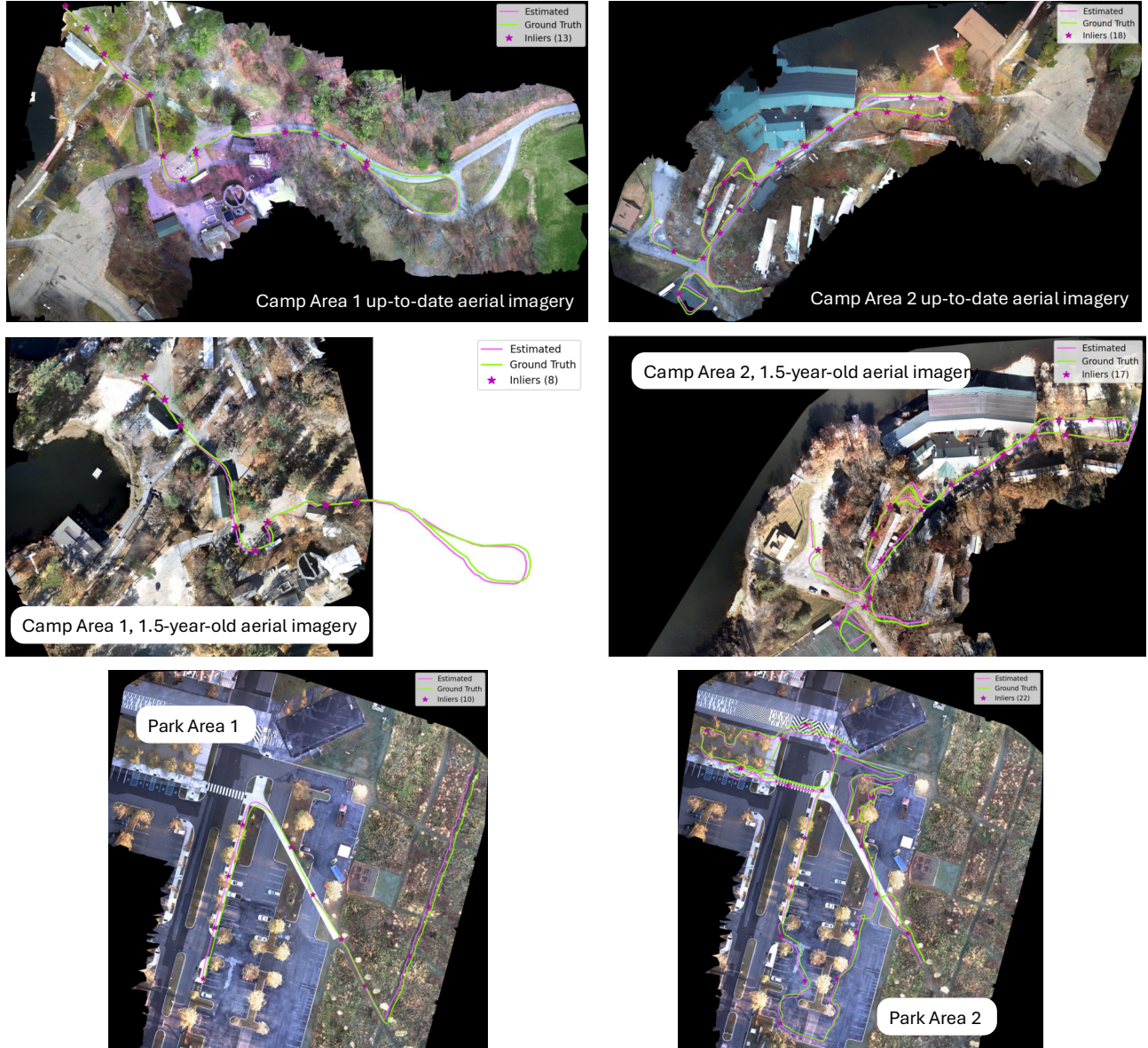


Fig. 5: Cross-view robot trajectory estimation in different environments. Inlier cross-view loop closures, shown with a purple star, are used to accurately estimate the pose of the robot at a sequence of submap centers. In each case, our method accurately estimates the robot geo-location over the course of the robot trajectory.

environments. First, we show that our system achieves competitive results in the widely used KITTI dataset [28], showcasing a large-scale urban environment. Depth data for the mapping pipeline is obtained by projecting LiDAR points into the front RGB camera frame. We use aerial imagery from Vexcel as references. Second, we test on two challenging self-collected datasets containing strong natural and unstructured features, shown in Fig. 5. The first self-collected dataset is in a park area containing a large meadow, park, and parking lot, and the second includes two different wilderness camp areas. Aerial imagery is collected onboard a drone with a downward-facing camera and stitched together into an orthomosaic. Ground

data is collected onboard two Clearpath Husky UGVs and two Boston Dynamics Spot robots. We use DLIO [29] for odometry onboard the Huskies and the built-in visio-kinematic odometry of the Spots. Each robot is equipped with an onboard ZED 2i camera which provides color and depth inputs to the mapping pipeline. Ground truth is obtained by fusing odometry (used to inform rotation) with GPS measurements. For each of these datasets except KITTI, mapping can be run faster than real-time at 6 Hz. Future work includes mapping speedups for KITTI. We show that in each of these environments, our algorithm is able to associate metric-semantic primitives across the two aerial/ground representations to estimate accurate

TABLE I: KITTI Absolute Trajectory Error, ATE (m)

	Method	Seq. 00	Seq. 02	Seq. 09
OSM	OSM-SLAM [18]	3.8	1.3	–
	Przewodowski [19]	2.8	3.1	3.6
	Li [20]*	1.6	fail	1.1
Aerial	Miller [4]	2.0	9.1	7.2
	Ankenbauer [17]	3.5	4.4	3.0
	Proposed	1.4	4.3	1.6

* Did not report whether error is reported as RMSE or mean or whether error is absolute in a global position sense or whether [30] was used to provide a least-squares alignment.

TABLE II: Trajectory Estimation Error (m) on Self-Collected Data

Area	Seq.	Length (m)	Trajectory Estimation Error (m)	
			Up-to-date	1.5-year-old
Camp Area	1	460	1.2	2.0
	2	760	1.5	3.2
	3	540	2.1	2.2
	4	1000	3.2	4.6
	5	1430	3.0	3.1
Park Area	1	280	1.9	1.9 [†]
	2	690	1.4	2.1 [†]
Mean±Std		–	2.0±0.7	2.7±0.9

[†] 1-year-old aerial imagery used for Park Area sequences.

poses.

We show the root-mean-squared absolute trajectory error (ATE) results on the KITTI dataset [28] in Table I, highlighting that the proposed method is competitive with other approaches for global pose estimation. The trajectory error is computed by taking the raw translation difference between the estimated position of the robot in UTM coordinates and the true UTM robot position from GPS. We note that the top rows of Table I rely on OSM to provide a prior for drivable area. While this is helpful in an autonomous driving scenario, we show that we are able to achieve low error with relaxed assumptions, relying only on an aerial overhead image. These relaxed assumptions are key for operating in less structured environments and when a mobile robot traverses areas not defined by roads, as demonstrated in our self-collected datasets.

In Table II we show that our algorithm transfers well to non-urban areas, achieving an average trajectory estimation error of 2.0 m across a total of over 5 km of traversal. We further challenge our method by testing it on out-of-date aerial imagery. The Camp Area dataset poses the additional challenge of localizing from an aerial image collected in the fall while the ground data is taken in the spring. Although the trajectory error increases when using an old aerial reference photo, the proposed algorithm is still able to achieve a low trajectory error of 2.7 m.

We also conduct an experiment, using the Camp Area dataset, to evaluate the per-submap registration success rate of our method, and report the results in Table III. Robot trajectories are split up into short traversals, each with a path length of 50 m, and given to the matching/registration module to estimate the pose of the robot at the center of the trajectory. Given n probability-ordered hypotheses, we evaluate whether

TABLE III: Registration Success Rate of Camp Area Submaps (<5 m, <5 deg)

Rotation	Method	Top-1	Top-5	Top-10	Top-25
Known	Points Only	36.2%	44.7%	51.1%	55.3%
	Points + Lines	42.6%	59.6%	72.3%	80.9%
Unknown	Points Only	4.3%	10.6%	17.0%	27.7%
	Points + Lines	14.9%	27.7%	38.3%	46.8%

any of the top n hypotheses is a pose estimate with less than 5 m translation error and less than 5 deg rotation error. Success rates at different n values are evaluated. In particular, we evaluate the utility of including lines in submap pose estimation, and compare matching points and lines against matching points only. In the more challenging case of unknown rotation, we show that using lines to inform trajectory estimation roughly doubles the success rate. This important capability enables accurate geographic pose estimation without any initial pose information.

V. CONCLUSION

We have demonstrated our cross-view localization pipeline on a range of both urban and natural environments and have shown the ability to associate primitives in a zero-shot manner for registering ground maps to aerial overhead images. One current limitation of the proposed work is a low recall rate from place recognition, requiring the algorithm to run matching on many non-overlapping ground submap and aerial patch pairs. While these can be rejected downstream by our robust pose graph optimization, scalability would be improved with higher cross-view place recognition recall. Furthermore, we have seen empirically that DINO feature similarity tends to be less discriminative across aerial and ground views than between two ground views. Future work includes investigating methods to bridge the semantic differences between ground and aerial views.

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